USN | | | | | | | | | | [22EEMS01]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER -2024

Introduction to Electric and Hybrid Vehicles

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define aerodynamic drag and write its formula	L1	CO2	**02**
	b.	Mention the different components of EV Drivetrain	L1	CO3	**02**
	c.	Explain the different types of sensors used for Electric Vehicles	L2	CO4	**02**
	d.	Which are the various factors considered for selecting the appropriate energy storage devices	L1	CO5	**02**
	e.	Broadly classify the energy management strategies used in hybrid and electric vehicles	L2	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Elaborate the environmental and social impacts of Hybrid Electric Vehicles.	L4	CO2	**05**
		OR			
	b.	With neat block diagram explain the power source categorization of electric vehicles.	L2	CO2	**05**
3.	**a.**	Illustrate the typical performance characteristics of ICE and EV	L3	CO3	**05**
		OR			
	b.	Classify the four HEV configurations with the help of block diagram representation	L2	CO3	**05**
4.	**a.**	Discuss in detail about the control of permanent magnet motor drives.	L3	CO4	**05**
		OR			
	b.	Write a short note on various methods of speed control of induction motors.	L3	CO4	**05**
5.	**a.**	Illustrate the need of hybrid storage devices in Electric Vehicles	L3	CO5	**05**
		OR			

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	b.	Differentiate fuel cells and batteries	L2	CO5	**05**
6.	**a.**	Elaborate the benefits of energy management	L4	CO6	**05**
		OR			
	b.	Discuss the fuzzy rule-based energy management strategy	L2	CO6	**05**
7.	**a.**	Discuss the properties of lithium-ion batteries	L2	CO5	**05**
		OR			
	b.	Illustrate the working of fuel cell with the neat constructional diagram	L3	CO5	**05**
		PART-C	**6X10=60**		
8.	**a.**	Briefly explain the history of hybrid electric vehicles	L2	CO2	**10**
		OR			
	b.	With the help of block diagrams Elaborate the different power flow control modes of a typical parallel hybrid system	L4	CO3	**10**
9.	**a.**	With a neat block diagram explain the concept of EV drivetrain	L2	CO3	**10**
		OR			
	b.	Illustrate the working of complex hybrid system with the help of block diagram	L3	CO3	**10**
10.	**a.**	What are the factors considered for selection of motor in EVs	L1	CO4	**10**
		OR			
	b.	Elaborate the different types of motors available for EV applications	L4	CO4	**10**
11.	**a.**	With a neat constructional diagram, explain the working principle of Nickel Cadmium battery. Also write the cell reaction during charging and discharging process	L3	CO4	**10**
		OR			
	b.	Discuss the Configuration and control of DC Motors	L3	CO4	**10**
12.	**a.**	Discuss the implementation issues of energy management strategies	L3	CO6	**10**

		OR			
	b.	Discuss the optimization-based Energy management strategies	L3	CO6	**10**
13.	**a.**	Discuss the role of ultracapacitors in EV	L3	CO5	**10**
		OR			
	b.	Summarize the following **i.** Advantages of FCEV over conventional energy resources **ii.** Challenges of FCEV **iii.** Advantages & disadvantages of FCEV **iv.** Applications of FCEV	L2	CO5	**10**